

Lab #03: RAM/ROM

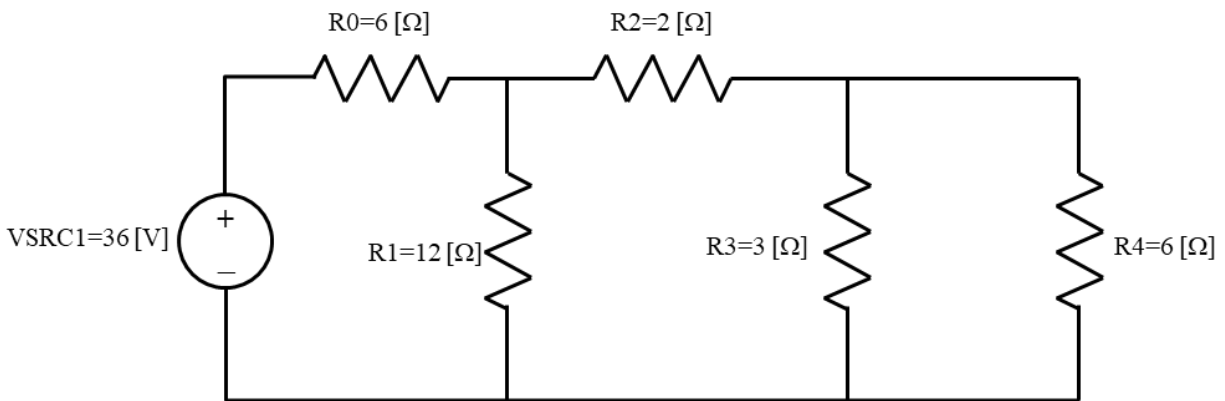
Overview

Download the starter project named ECE_Lab3_Team##.zip from Teams/Canvas.

Refer to the following circuit schematic to complete the programming tasks. You have completed this lab earlier in the semester but using only registers. Now you are going to do it using RAM and ROM.

IMPORTANT:

- You are not allowed to use immediate values in your calculations. You MUST use the results in ROM/RAM from previous calculations to calculate the next results.
- Place the code for each task in the designated area in the main.s file.



TASK 1

- Create a read-only data area for your constants called **MyConstants**. Align the start of that section to a memory address that is a multiple of 4.
- Create a 16-bit constant named **VSRC1** and initialize it to the value 36.
- Create 8-bit constants named **Resistor0**, **Resistor1**, **Resistor2**, **Resistor3**, & **Resistor4** and initialize them with the values shown in the circuit above.
- Align the end of that section to a memory address that is a multiple of 4.

TASK 2

- Create a read/write data area for your variables called **MyVariables**. Align the start of that section to a memory address that is a multiple of 4.

- b) Create 16-bit variables named **ISRC1**, **VSRC2**, **ISRC2**, **VSRC3** and **IR4**.
- c) Create 8-bit variables named **Resistor5**, **Resistor6**, **Resistor7**.
- d) Align the end of the section to a memory address that is a multiple of 4.
- e) Create a read-only code area for your program called `|.text|`. Align the start of that section to a memory address that is a multiple of 4.

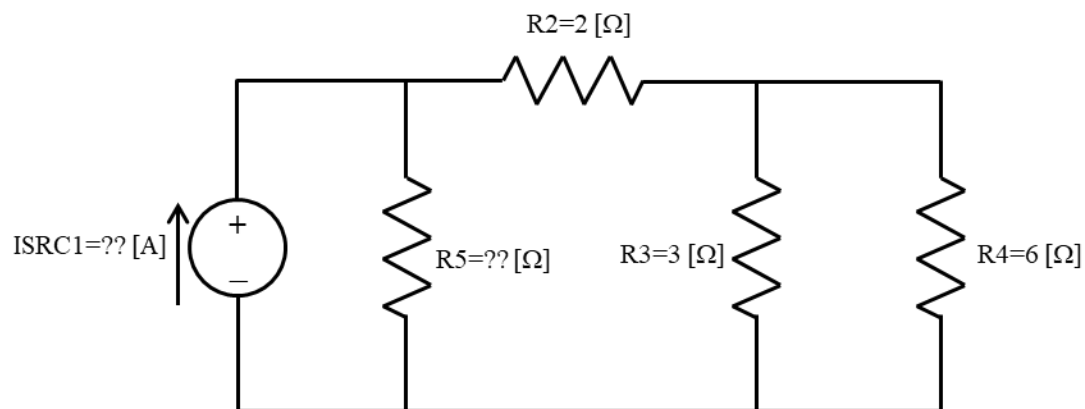
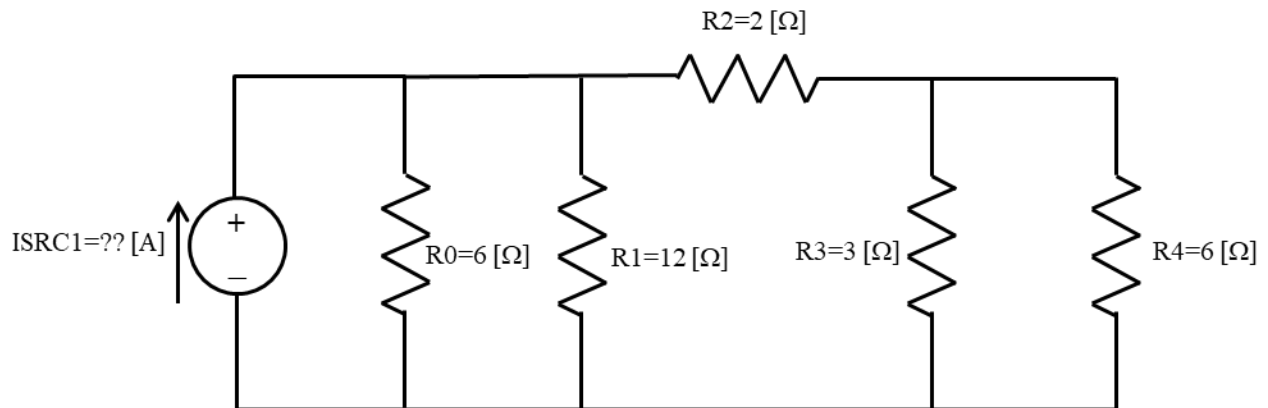
TASK 3

The first diagram below shows the circuit after a source transformation on the original circuit.

The second, after finding an equivalent resistance.

Calculate the value of **ISRC1** ($=VSRC1/R0$) and place the value in the variable **ISRC1**.

Calculate the value of $R5$ in ohms and place the value in the variable **Resistor5**.

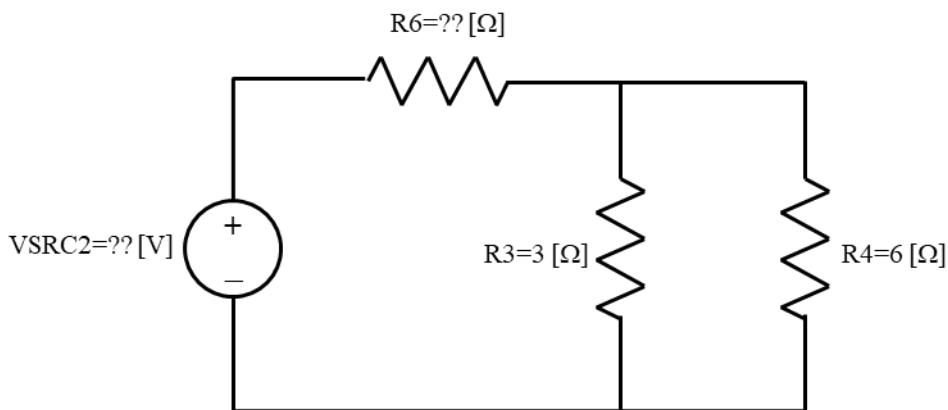
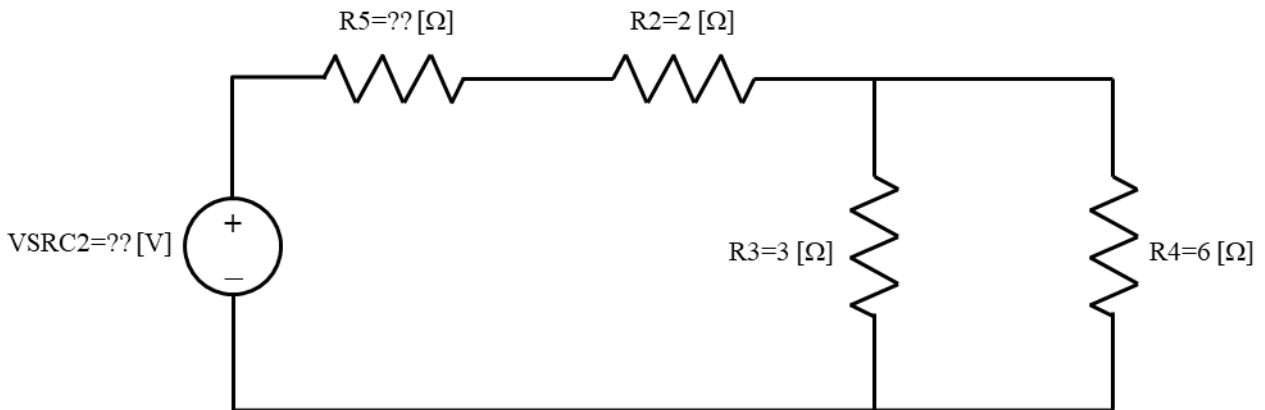


TASK 4

The following two diagrams show the circuit after a source transformation and finding an equivalent resistance.

Calculate the value of **VSRC2** ($=ISRC1 \cdot R5$) and place the value in the variable **VSRC2**.

Calculate the value of R6 in ohms and place the value in the variable **Resistor06**.

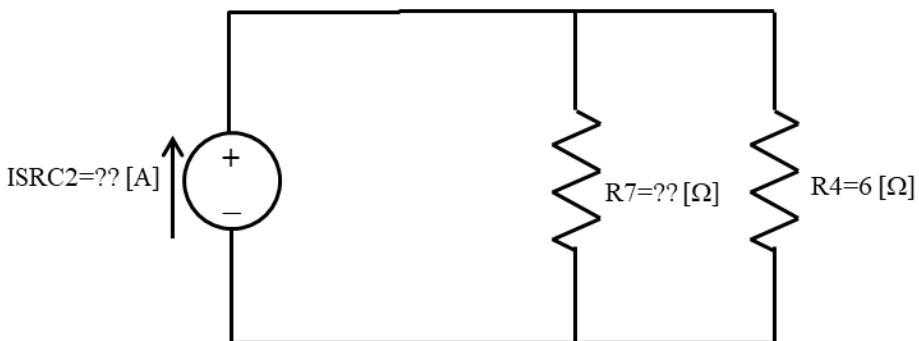
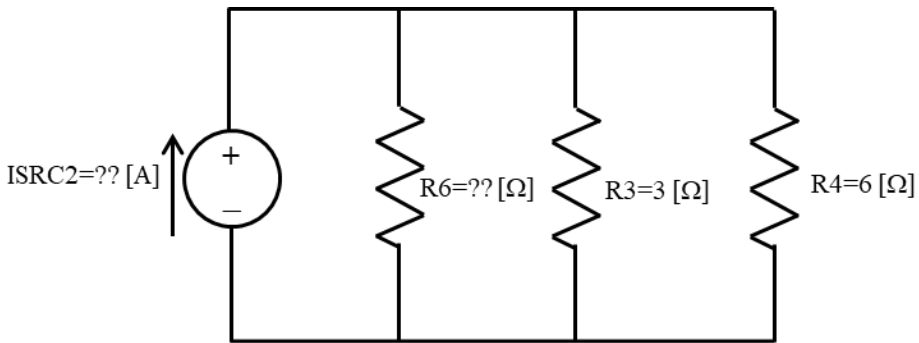


TASK 5

The following two diagrams show the circuit after a source transformation and finding an equivalent resistance.

Calculate the value of **ISRC2** ($=V_{SRC2}/R6$) and place the value in the variable **ISRC2**.

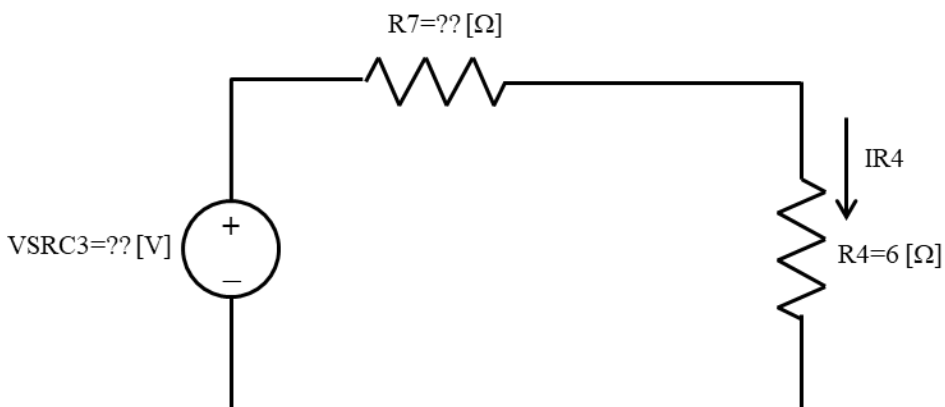
Calculate the value of R7 in ohms and place the value in the variable **Resistor7**.



TASK 6

The following two diagrams show the circuit after a source transformation and finding an equivalent resistance.

Calculate the value of **VSRC3** ($= ISRC2 * R7$) and place the value in the variable **VSRC3**.
Calculate the current $IR4$ and place the result in the variable **IR4**.



TASK 7

Change the value of the constant VSRC1 to 72 and verify that your answers are still correct.

Submission

Submit the complete and working project to Canvas as one zip file.

- The project file will be named Lab3.uvprojx
- The project folder will be named ECE_3436_Lab3_Team##.
- Therefore, the zip file will be named ECE_3436_Lab3_Team##.zip.